# Worksheet 26 · Straight-Line Equations IGCSE

*Edexcel IGCSE Mathematics · past paper questions*

| Name: | Subject: Math | Date: |
|---|---|---|
| **Learning Objectives:**<br>Students will be able to:<br>• Find the gradient of a line between two points<br>• Find the equation of a straight line in the form y = mx + c<br>• Find midpoints and lengths of line segments | | |
| **Grade:** / 42 | **Teacher's Feedback:** | |

*Answer in the space given. Show your working. Marks for each part are shown in brackets.*

## Questions *Straight-Line Equations*

### Question 1 (4 marks)

The point $A$ has coordinates $(5, -4)$
The point $B$ has coordinates $(13, 1)$

(a) Work out the coordinates of the midpoint of $AB$.

( ..... , ..... )

(2)

Line **L** has equation $y = 2 - 3x$

(b) Write down the gradient of line **L**.

(1)

Line **L** has equation $y = 2 - 3x$

(c) Does the point with coordinates $(100, -302)$ lie on line **L**?
You must give a reason for your answer.

.....

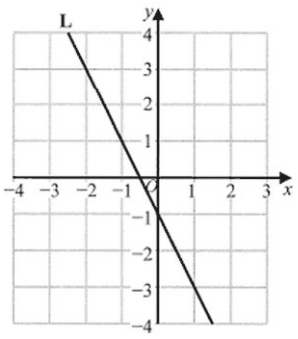
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(1)

### Question 2 (3 marks)

Line **L** is drawn on the grid.

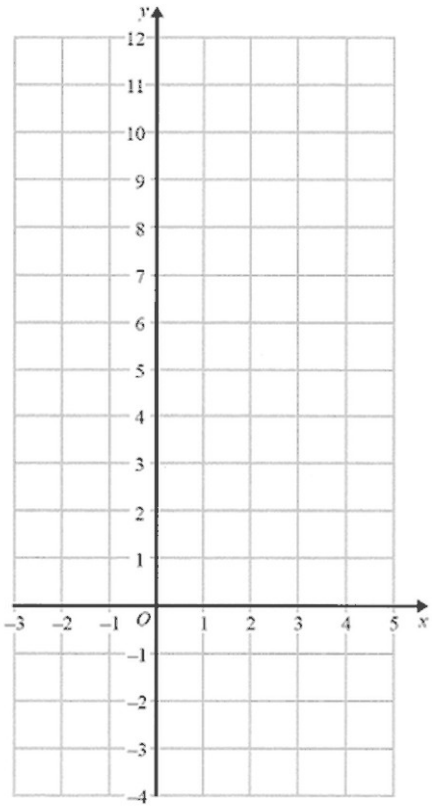



Find an equation for **L**.

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### Question 3 (4 marks)

On the grid, draw the graph of $y + 2x = 6$ for values of $x$ from $-2$ to $4$.



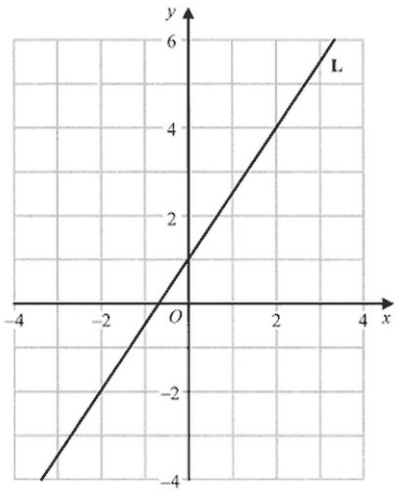

### Question 4 (2 marks)

Find the gradient of the straight line with equation $5x + 2y = 7$

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## Question 5 (3 marks)

The line **L** is drawn on the grid.




Find an equation for **L**.

.....

## Question 6 (2 marks)

Point $A$ has coordinates $(-3, 11)$
Point $B$ has coordinates $(47, b)$
The midpoint of $AB$ has coordinates $(a, -19)$

Find the value of $a$ and the value of $b$.

$a =$ .....

$b =$ .....

## Question 7 (5 marks)

Here are the equations of four straight lines.

| | | |
|---|---|---|
| Line **A** | $y$ | $= 2x + 3$ |
| Line **B** | $2y$ | $= 6 - 3x$ |
| Line **C** | $4x - 2y$ | $= 3$ |
| Line **D** | $y$ | $= 3 - 2x$ |

Two of these lines are parallel.

(*a*) Which two lines?

.....

**(2)**

Line **L** has a gradient of $-\frac{5}{2}$ and passes through the point with coordinates $(1, 3)$

(*b*) Find an equation of **L**.
Give your answer in the form $ax + by = c$ where $a$, $b$ and $c$ are integers.

.....

**(3)**

## Question 8 (2 marks)

Write down an equation of the straight line with gradient $-3$ and which passes through the point with coordinates $(0, 5)$

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(2)

## Question 9 (3 marks)

The straight line **L** has equation $2y + 7x = 10$

(a) Find the gradient of **L**

(2)

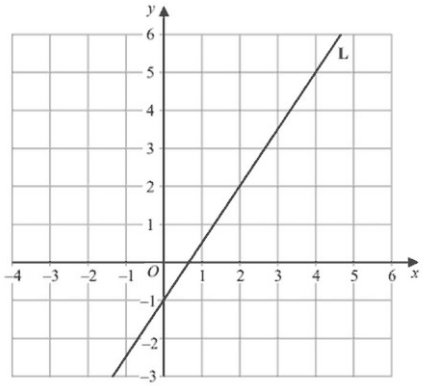
(b) Find the coordinates of the point where **L** crosses the $y$-axis.

(.....)

(1)

## Question 10 (3 marks)

Line **L** is drawn on the grid.




Find an equation for **L**
Give your answer in the form $y = mx + c$

.....

## Question 11 (4 marks)

The points $(1, -1)$ and $(4, 7)$ lie on the straight line **L**.

Find an equation for **L**.
Give your equation in the form $ax + by = c$ where $a$, $b$ and $c$ are integers.

.....

## Question 12 (4 marks)

Find an equation of the line that passes through the points $(-3, 5)$ and $(1, 2)$
Give your answer in the form $ax + by = c$ where $a$, $b$ and $c$ are integers.

.....

(4)

**Question 13** **(3 marks)**

Line **A** has equation $3x - 4y = 5$

Line **B** goes through the points (4, 7) and (–1, 3)

Are lines **A** and **B** parallel?

Show your working clearly.